Kyle Han

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Education

Stony Brook University — MS in Computer Engineering | (GPA 3.56/4) **Aug 2024 – May 2025**

- Relevant Coursework: Computer Architecture, Parallel Processing Architectures, Adv. VLSI System Design, Adv. Digital Sys. Design

Stony Brook University — BE in Electrical Engineering | Magna Cum Laude (GPA 3.8/4) **Aug 2020 – May 2024**

- Relevant Coursework: Computer Architecture, Real-time Operating Systems, Design of Secure IOT Systems.
- Organizations: IEEE (Secretary, Graduate Advisor), Robotics, Motorsports (Electrical Subsystem), Tau Beta Pi.

Projects

Dual Issue 11–Stage Superscalar Multimedia Unit (SONY Cell SPU Model) — *SystemVerilog, Siemens Questasim, C++*

- Built an RTL model for a dual–issued, 11–stage SIMD processor with 7 execution units based on Sony's Cell SPU.
- Designed forwarding logic and a stalling hazard detection unit to resolve RAW, WAW, control, and structural hazards.
- Wrote a bash script to automate design validation using a directed testbench with file–driven initialization.
- Implemented a 2-pass C++ assembler supporting labels and inline comments.

Matrix Multiplication Accelerator Hardware Accelerator — *SystemVerilog, Siemens Questasim, Synopsys Design Compiler*

- Designed and synthesized an RTL Matrix Multiplication Unit with parallel pipelined components (MAC unit, FIFO, control logic).
- Implemented an AXI4–Stream with the TUSER sideband to support arbitrary matrix sizes and blocked matrices.
- Improved throughput and reduced stalls by 50% using input buffering and deeper FIFOs.
- Validated RTL design using a constrained–random SystemVerilog testbench and DPI-C reference model.

Pipelined Synchronous CMOS 8-bit Carry Select Adder — *Cadence Virtuoso, HSPICE*

- Designed a custom pipelined 8–bit carry select adder using FreePDK45 design rules, targeting 4 GHz and low power (<1mW).
- Laid out 39 flip–flops, 10 muxes, and 16 full adders on a 21.7µm x 37.5µm die without using standard cells.
- Verified timing and accuracy by performing post–layout simulation in HSPICE after passing DRC and LVS.

GPGPU Based Distributed Matrix Multiplier & Parallel Pathfinder — C, *CUDA, MPI, OpenMP, PThreads, ISPC, CUDA-GBD*

- Achieved a 9,774× acceleration in matrix–multiplication by implementing a shared-memory tiled CUDA kernel and distributing computation across 4 GPUs on 2 nodes via MPI, minimizing data transfer overhead.
- Developed CUDA kernels for global two–minimia identification using block–level reduction and atomic operations.
- Applied a CUDA–parallelized Dijkstra implementation to compute the shortest path between the identified minimas.

Experience

IEEE Northeast Regional Student Conference — *Conference Chair* **Jan 2025 – May 2025**

- Organized a sold–out 3–day conference for 300+ attendees by coordinating staffing, logistics, and operations.
- Secured $44,000 in support for 30 events by drafting grant proposals and negotiating partnerships with 17+ orgs.

Zebra Technologies — *Supply Chain and Repair Engineering Intern* **May 2024 – Aug 2024**

- Accelerated repair of Zebra Mobile Computers by 5% by writing C++ firmware for an automated triage system.
- Improved shipping cost accuracy by designing a Databricks forecasting model with a self–correcting feedback system using SQL and Google Cloud; visualized model performance using Power BI.

TC Electric — *Innovation Intern* **May 2023 – Jan 2024**

- Eliminated 60% of redundant fields in required MTA subcontractor forms, by building validated JavaScript web forms and Power Automate, reducing potential delays due to incorrect information.
- Identified limitations in existing clearance verification practices, and through a series of weekly innovation reports, recommended a 3D LiDAR scanning solution, which the company adopted as a modernization initiative.

ITC Midwest — *Line Design Intern* **May 2022 – Aug 2022**

- Located 5 grid hardening opportunities by analyzing 13 years of outage data with Python, assessing galloping behavior to propose future infrastructure improvement projects.
- Modernized 40 miles of microfiched plan and profiles using PLS-CADD, providing field engineers with accurate documentation.

Skills

CAD: Altium Designer, Cadence OrCAD, Autodesk Fusion 360, Autodesk Inventor
EDA: Synopsys Design Compiler, Synplify, Lattice Diamond, Siemens QuestaSim, Xilinx Vivado, Cadence Virtuoso, LTSpice, HSPICE
Programming: C/C++, CUDA, Python, AVR/MIPS Assembly, ISPC, SystemVerilog, VHDL, MATLAB, Git
Protocols: AMBA AXI4-Stream, I2C, SPI, UART

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